6 Advancements in Information and Communication Technologies

Artificial Intelligence

- It is a branch of computer science focussing on creating machines, which can engage in behaviours that humans consider intelligent.
- It is a combination of computer science, physiology, and philosophy.
- Artificial intelligence includes:
  - Games such as chess, video games, etc.
  - Expert systems: coding machines to make judgments in practical situations
  - Natural language
  - Neutral networks
  - Robotics

Evolved EDGE Technology

EDGE (Enhanced Data Rates for GSM Evolution)

It is an enhancement for 2.5 G GSM/GPRS networks. EDGE can function on any network with GPRS enabled on it.

Evolved EDGE

It is a modified and upgraded version of EDGE, where latencies are reduced by lowering the transmission time by half. Bit transfer rates are increased up to 1 Mbit/second. Signal quality is also improved by using dual antennas, which improve average bit rates and spectrum efficiency.

Holographic Storage

- It is a computer storage that uses laser beams to store computer data in three dimension and hence achieves move data storage than conventional optical and magnetic storage.
- The holographic method can be compared with the holographic sticker, which can display three-dimensional objects, even though the sticker is only two-dimensional. The idea of holographic storage is to use this type of technology to store computer information. This can store large amounts of data in a small space.
- Magnetic and optical data storage devices (computer hard drive and CDROM) records individual bits (a bit at a time) in a linear fashion and stored them as distinct magnetic or optical changes on the surface of the magnetic tape.
- Holographic storage records and reads millions of bits in parallel on the same areas of the medium by utilising light at different angles. Holographic storage enables faster data transfer than optical storage.

Holographic Versatile Disc (HVD)

It is an optical disc technology that uses holographic storage and can store 1–10 terabytes of data. No company has yet released an HVD.

Hyper Text Mark up Language (HTML 5.0)

- Hyper is the opposite of linear. Earlier, the computer programme ran linearly, when the programme had executed one action, it went to the next line and after that the next line, and so on. But HTML 5.0 is different, you can go where ever you want.
- Mark up is what you do with the text. You are marking up the text the same way you do in a text editing programme with headings, bullets, bold text, etc.
- HTML was invented by Tim Berners Lee. HTML is the main markup language for web pages.
- HTML elements form the building blocks of all websites. HTML allows images and objects to be embedded and can be used to create interactive forms.

Quantum Computing

- It employs properties described by quantum mechanics to represent data and perform computations. A quantum computer is a device that employs properties described by quantum mechanics to represent data and perform computations.
- Elementary particles such as protons, neutrons, and electrons can persist in two or more states at the same time. That principally

makes them to be harnessed as memory and processing units.

- According to quantum mechanics, an electron has a binary property, known as 'spin.' This spin can be used to represent data, just like transistors, which are used in conventional computers to represent digital data '1' or '0' by their output voltage.
- The basic unit of quantum computing is the 'quit' (quantum + bit). While the bit in conventional computers is either a digital '1' or ' 0', the qubit can contain both values '1' and '0' simultaneously. Hence, any operations performed on a single qubit effectively acts on both values at the same time. Therefore, by performing a single operation on a single qubit, we have performed two operations. Therefore, a two qubit system would perform four operations.
- It can quickly and easily perform calculations that are incredibly time consuming on conventional computers.
- It will bring down the space needed to store the data and will increase the speed at which data can be processed.
- Hence quantum computing can be more highly efficient than conventional computing for processing the same amount of data and hence will consume minimum energy.
    - Quantum computers can replace silicon chips.
    - Quantum computing is not well suited for tasks such as word processing and email, but it is ideal for tasks such as cryptography and indexing a very large database.

Stuxnet

Stuxnet is a computer worm discovered in July 2010. It targeted five Iranian organisations, with the probable target, widely suspected to be uranium enrichment infrastructure in Iran.

Big Data

- Big data refers to the vast amount of unstructured, semistructured, and structured data, which have the ability to be processed for insights that overshoot a business on a daily basis.
- It is distinguished by the three 'V's: volume, variety, and velocity of the data processed.
- Typically, big data consist of data files whose sizes exceed the capacity of frequently applied tools to acquire, analyse, organise, and process information within a reasonable timeframe.

Net Neutrality

- Net neutrality is the principle that individuals should be free to access all content and applications equally, regardless of the source without internet service provider's discrimination against specific online services or websites.
- In other words, net neutrality (also known as network neutrality) is the principle that internet service providers and governments should treat all data on the Internet the same, by not discriminating or charging differentially for user, content, site, etc.

Practice Exercise

1. Consider the following statements.

I. The qubits of the quantum computer are fundamentally the same as the bits of a classical computer.

II. The qubits of a quantum computer can hold more information than their classical counterpart bits.

Which of the statements given above is/are correct?

(A) Only I
(B) Only II
(C) Both I and II
(D) Neither I nor II

2. Artificial intelligence is:

(A) The embodiment of human intellectual capabilities within a computer.
(B) A set of computer programmes that produce output that would be considered to reflect intelligence as if it were generated by humans.
(C) The study of mental faculties through the use of mental models implemented on a computer.
(D) All of the above

3. What is artificial intelligence?

(A) Putting your intelligence into computer
(B) Programming with our intelligence
(C) Making a machine intelligent
(D) Putting more memory into the computer

4. Consider the following statements regarding Holostore.

I. It is a device that reads and writes data in an optical form.

II. It is a computer storage device.

III. It refers to Institutions where holography is taught.

Which of the above statements are correct?

(A) I, II, and III
(B) I and III only
(C) I and II only
(D) II and III only

5. Which particular generation of computers is associated with artificial intelligence?

(A) Fourth
(B) Second
(C) Third
(D) Fifth

Answer Key

1. (C) **2.** (D) **3.** (C) **4.** (C) **5.** (D)